

## LAB 3- Detecting Anomalies with Prophet

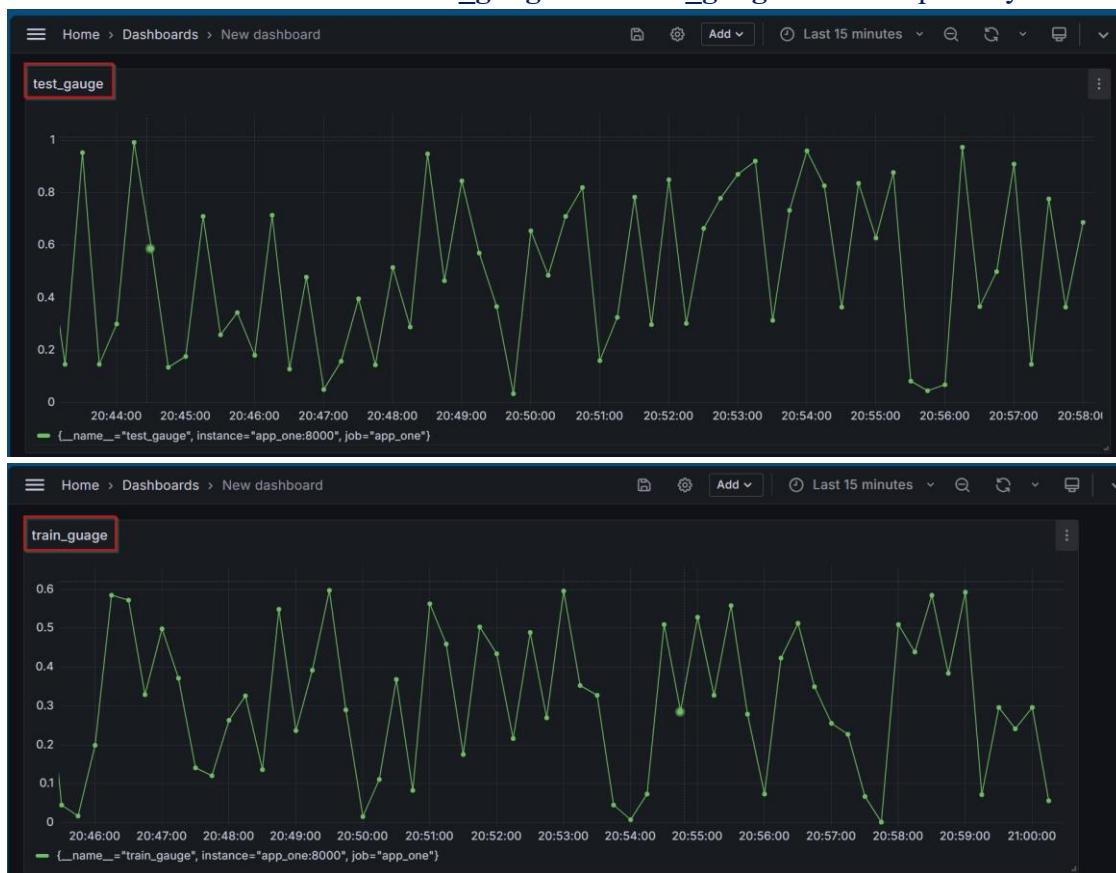
Confirming the app.py for app\_one is still the same.

```
app.py U X
LAB3-Detecting_Anomalies_with_Prophet > containers > app_one > app.py
1 from prometheus_client import start_http_server, Gauge, Histogram
2 import random
3 import time
4
5 # Define metrics
6 test_gauge = Gauge('test_gauge', 'Test Gauge between 0 and 1')
7 train_gauge = Gauge('train_gauge', 'Train Gauge between 0 and 0.6')
8 test_hist = Histogram('test_hist', 'Test Histogram between 0 and 1', buckets=(0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0))
9 train_hist = Histogram('train_hist', 'Train Histogram between 0 and 0.6', buckets=(0.1, 0.2, 0.3, 0.4, 0.5, 0.6))
10
11 def emit_data():
12     """Emit fake data"""
13     time.sleep(5)
14     test_value = random.uniform(0, 1)
15     train_value = random.uniform(0, 0.6)
16     test_gauge.set(test_value)
17     train_gauge.set(train_value)
18
19     test_hist.observe(test_value)
20     train_hist.observe(train_value)
21
22
23
24 if __name__ == '__main__':
25     start_http_server(8000)
26     while True:
27         emit_data()
```

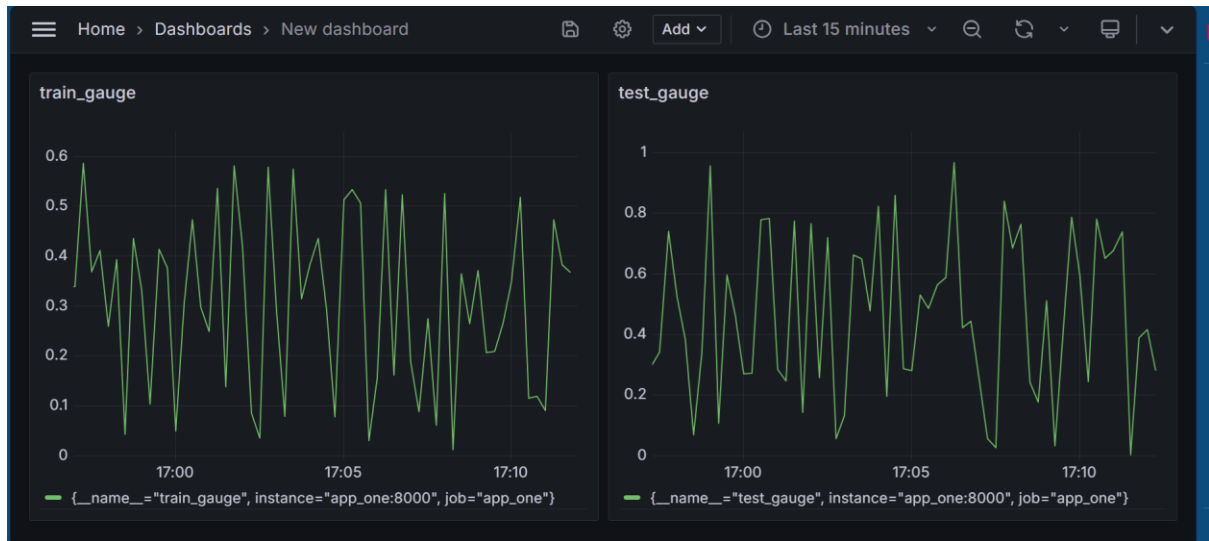
Revisited the app\_one from Lab 2 and created the containers with **docker compose up -d**

```
oadewusi@oluSec:~/aiops/Assignments/LAB3-Detecting_Anomalies_with_Prophet$ docker compose up -d
[+] Running 7/7
 ✓ Container prometheus_node_exporter Started
 ✓ Container prometheus Started
 ✓ Container app_one Started
 ✓ Container prometheus_push_gateway Started
 ✓ Container postgres Started
 ✓ Container postgres_exporter Started
 ✓ Container grafana Started
```

Screenshots on Grafana for the **test\_gauge** and **train\_gauge** metrics separately



The metrics together on the dashboard:



Prepared a python application called **monitor\_app** that reads the training data metric for a period of time, builds a Prophet model from it, and evaluates the model against the test metric. Simply print the anomalies detected to the console log.

```
monitor_app.py X
LAB3-Detecting Anomalies with Prophet > containers > monitor_app > monitor_app.py
1 import pandas as pd
2 from prophet import Prophet
3 from prometheus_client import Gauge, start_http_server
4 from sklearn.metrics import mean_absolute_error, mean_absolute_percentage_error
5 import requests
6 import numpy
7 from datetime import datetime, timedelta
8 import logging
9 import time
10
11 logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
12 logging.getLogger('prophet').setLevel(logging.WARNING)
13 logging.getLogger('candstanpy').setLevel(logging.WARNING)
14
15 anomaly_count, mae_metric, mape_metric = Gauge('anomaly_count', 'Number of anomalies detected'), Gauge('mae_metric', 'Mean Absolute Error'), Gauge('mape_metric', 'Mean Absolute Percentage Error')
16 start_http_server(8000)
17
18 def fetch_metric_data(metric_name, start_time, end_time, step='15s'):
19     url, params = f"http://prometheus:9090/api/v1/query_range", {'query': metric_name, 'start': start_time.timestamp(), 'end': end_time.timestamp(), 'step': step}
20     results = requests.get(url, params=params).json().get('data', {}).get('result', [])
21     if not results: return pd.DataFrame()
22     df = pd.DataFrame(results[0]['values'], columns=['ds', 'y'])
23     df['ds'] = df['y'] = pd.to_datetime(df['ds'], unit='s').dt.strftime('%Y-%m-%d %H:%M:%S'), df['y'].astype(float)
24     return df
25
26 def train_and_detect(train_data, test_data):
27     model = Prophet(growth='flat', daily_seasonality=False, weekly_seasonality=False, yearly_seasonality=False)
28     model.fit(train_data)
29     forecast = model.predict(test_data)
30     merged = forecast.merge(test_data, on='ds', how='inner')
31     return merged[merged['y'] > merged['yhat_upper']], forecast
32
33 def main():
34     all_results, cumulative_anomaly_count = pd.DataFrame(), 0
35     while True:
36         try:
37             end_time = datetime.now()
38             train_data = fetch_metric_data('train_gauge', end_time - timedelta(minutes=5), end_time)
39             if train_data.empty:
40                 logging.warning("Training data empty, retrying in 60s")
41                 time.sleep(60)
42                 continue
43             time.sleep(60)
44             test_data = fetch_metric_data('test_gauge', end_time, datetime.now())
45             if test_data.empty:
46                 logging.warning("Test data empty, skipping cycle")
47                 continue
48
49             anomalies, forecast = train_and_detect(train_data, test_data)
50             cumulative_anomaly_count += len(anomalies)
51             anomaly_count.set(cumulative_anomaly_count)
52
53             mae = mean_absolute_error(test_data['y'], forecast['yhat'][:len(test_data)])
54             mape = mean_absolute_percentage_error(test_data['y'], forecast['yhat'][:len(test_data)])
55             mae_metric.set(mae)
56             mape_metric.set(mape)
57
58             current_results = forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].copy()
59             current_results['y'], current_results['timestamp'], current_results['anomaly_count'] = test_data['y'].values, datetime.now().strftime('%Y-%m-%d %H:%M:%S'), cumulative_anomaly_count
60             current_results = current_results[['timestamp', 'ds', 'y', 'yhat', 'yhat_lower', 'yhat_upper', 'anomaly_count']]
61             all_results = pd.concat([all_results, current_results], ignore_index=True)
62
63             logging.info(f"Anomalies: {len(anomalies)}, Cumulative: {cumulative_anomaly_count}, MAE: {mae:.4f}, MAPE: {mape:.4f}")
64             pd.set_option('display.max_rows', None)
65             pd.set_option('display.max_columns', None)
66             pd.set_option('display.width', None)
67             logging.info("All Results:\n" + all_results.to_string(index=False))
68         except Exception as e:
69             logging.error(f"An error occurred: {e}")
70             time.sleep(60)
71
72 if __name__ == '__main__':
73     main()
```

Now packaging the application as a docker image:

Created a Dockerfile:

```
Dockerfile U X
LAB3-Detecting_Anomalies_with_Prophet > containers > monitor_app > Dockerfile > ...
1 FROM python:3.9-slim
2 WORKDIR /app
3 COPY . /app
4 RUN pip install --no-cache-dir -r requirements.txt
5 EXPOSE 8003
6 ENV PYTHONBUFFERED=1
7 CMD ["python", "monitor_app.py"]
8 ✨
9
```

With the requirements:

```
requirements.txt U X
LAB3-Detecting_Anomalies_with_Prophet > containers > monitor_app > requirements.txt
1 pandas
2 prophet
3 prometheus_client
4 scikit-learn
5 numpy==1.26.0
6 requests
7
```

Added the monitor\_app to the prometheus configuration file on port 8003:

LAB3-Detecting\_Anomalies\_with\_Prophet > containers > prometheus > ! config.yml

```
1  global:
2    scrape_interval: "5s"
3    evaluation_interval: "5s"
4
5  scrape_configs:
6    - job_name: "node_exporter"
7      static_configs:
8        - targets:
9          - "node_exporter:9100"
10   - job_name: "postgres_exporter"
11     static_configs:
12       - targets:
13         - "postgres_exporter:9187"
14   - job_name: "push_gateway"
15     static_configs:
16       - targets:
17         - "push_gateway:9091"
18   - job_name: "app_one"
19     static_configs:
20       - targets:
21         - "app_one:8000"
22   - job_name: "monitor_app"
23     static_configs:
24       - targets:
25         - "monitor_app:8003"
```

Also added the monitor\_app to the docker compose file on the port 8003

```
LAB3-Detecting_Anomalies_with_Prophet > docker-compose.yml
1  services:
2    app_one:
3      container_name: "app_one"
4      build: "./containers/app_one"
5      ports:
6        - "8000:8000"
7
8    app_two:
9      container_name: "app_two"
10     build: "./containers/app_two"
11     ports:
12       - "8001:8001"
13     environment:
14       DOCKER_NETWORK: push_gateway:9091
15
16    prometheus:
17      container_name: "prometheus"
18      build: "./containers/prometheus"
19      ports:
20        - "9090:9090"
21
22    grafana:
23      container_name: "grafana"
24      build: "./containers/grafana"
25      ports:
26        - "3000:3000"
27
28    node_exporter:
29      container_name: "prometheus_node_exporter"
30      image: "quay.io/prometheus/node-exporter"
31      ports:
32        - "9100:9100"
33
34    push_gateway:
35      container_name: "prometheus_push_gateway"
36      image: "prom/pushgateway"
37      ports:
38        - "9091:9091"
39
40    postgres:
41      container_name: "postgres"
42      image: postgres:13.3
43      restart: always
44      ports:
45        - "5432:5432"
46      environment:
47        POSTGRES_PASSWORD: example
48
49    postgres_exporter:
50      container_name: "postgres_exporter"
51      image: "wrouesnel/postgres_exporter"
52      ports:
53        - "9187:9187"
54      environment:
55        DATA_SOURCE_NAME: "postgresql://postgres:example@postgres:5432/postgres?sslmode=disable"
56      depends_on:
57        - postgres
58
59    monitor_app:
60      container_name: "monitor_app"
61      build: "./containers/monitor_app"
62      ports:
63        - "8003:8003"
```

Then readjusted the monitor\_app for the docker container:

```
monitor_app.py X
LAB3-Detecting-Anomalies_with_Prophet > containers > monitor_app > monitor_app.py
1 import pandas as pd
2 from prophet import Prophet
3 from prometheus_client import Gauge, start_http_server
4 from sklearn.metrics import mean_absolute_error, mean_absolute_percentage_error
5 import requests
6 import numpy
7 from datetime import datetime, timedelta
8 import logging
9 import time
10
11 logging.basicConfig(level=logging.INFO, format='%asctimes - %(levelname)s - %(message)s')
12 logging.getLogger('prophet').setLevel(logging.WARNING)
13 logging.getLogger('cndstanpy').setLevel(logging.WARNING)
14
15 anomaly_count, mae_metric, mape_metric = Gauge('anomaly_count', 'Number of anomalies detected'), Gauge('mae_metric', 'Mean Absolute Error'), Gauge('mape_metric', 'Mean Absolute Percentage Error')
16 start_http_server(8003)
17
18 def fetch_metric_data(metric_name, start_time, end_time, step='15s'):
19     url, params = f'http://prometheus:9090/api/v1/query_range', {'query': metric_name, 'start': start_time.timestamp(), 'end': end_time.timestamp(), 'step': step}
20     results = requests.get(url, params=params).json().get('data', []).get('result', [])
21     if not results: return pd.DataFrame()
22     df = pd.DataFrame(results[0]['values'], columns=['ds', 'y'])
23     df['ds'], df['y'] = pd.to_datetime(df['ds'], unit='s').dt.strftime('%Y-%m-%d %H:%M:%S'), df['y'].astype(float)
24     return df
25
26 def train_and_detect(train_data, test_data):
27     model = Prophet(growth='flat', daily_seasonality=False, weekly_seasonality=False, yearly_seasonality=False)
28     model.fit(train_data)
29     forecast = model.predict(test_data)
30     merged = forecast.merge(test_data, on='ds', how='inner')
31     return merged[merged['y'] > merged['yhat_upper']], forecast
32
33 def main():
34     all_results, cumulative_anomaly_count = pd.DataFrame(), 0
35     while True:
36         try:
37             end_time = datetime.now()
38             train_data = fetch_metric_data('train_gauge', end_time - timedelta(minutes=5), end_time)
39             if train_data.empty:
40                 logging.warning("Training data empty, retrying in 60s")
41                 time.sleep(60)
42                 continue
43             time.sleep(60)
44             test_data = fetch_metric_data('test_gauge', end_time, datetime.now())
45             if test_data.empty:
46                 logging.warning("Test data empty, skipping cycle")
47                 continue
48
49             anomalies, forecast = train_and_detect(train_data, test_data)
50             cumulative_anomaly_count += len(anomalies)
51             anomaly_count.set(cumulative_anomaly_count)
52
53             mae = mean_absolute_error(test_data['y'], forecast['yhat'][:len(test_data)])
54             mape = mean_absolute_percentage_error(test_data['y'], forecast['yhat'][:len(test_data)])
55             mae_metric.set(mae)
56             mape_metric.set(mape)
57
58             current_results = forecast[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].copy()
59             current_results['y'], current_results['timestamp'] = test_data['y'].values, datetime.now().strftime('%Y-%m-%d %H:%M:%S')
60             current_results['anomaly_count'] = test_data['y'].values, datetime.now().strftime('%Y-%m-%d %H:%M:%S'), cumulative_anomaly_count
61             current_results = current_results[['timestamp', 'ds', 'y', 'yhat', 'yhat_lower', 'yhat_upper', 'anomaly_count']]
62             all_results = pd.concat([all_results, current_results], ignore_index=True)
63
64             logging.info(f"Anomalies: {len(anomalies)}, Cumulative: {cumulative_anomaly_count}, MAE: {mae:.4f}, MAPE: {mape:.4f}")
65             pd.set_option('display.max_rows', None)
66             pd.set_option('display.max_columns', None)
67             pd.set_option('display.width', None)
68             logging.info(f"All Results:\n" + all_results.to_string(index=False))
69         except Exception as e:
70             logging.error(f"An error occurred: {e}")
71             time.sleep(60)
72
73 if __name__ == "__main__":
74     main()
```

Anomaly count, Mae and Mape metrics added to the application:

```
anomaly_count, mae_metric, mape_metric = Gauge('anomaly_count', 'Number of anomalies detected'),
Gauge('mae_metric', 'Mean Absolute Error'),
Gauge('mape_metric', 'Mean Absolute Percentage Error')
start_http_server(8003)

mae = mean_absolute_error(test_data['y'], forecast['yhat'][:len(test_data)])
mape = mean_absolute_percentage_error(test_data['y'], forecast['yhat'][:len(test_data)])
mae_metric.set(mae)
mape_metric.set(mape)
```

Then restarted the docker compose.

```
oadewusi@oluSec:~/aiops/Assignments/LAB3-Detecting-Anomalies_with_Prophet/containers/monitor_app$ docker compose up
[+] Running 9/0
  ✓ Container prometheus_push_gateway Created
  ✓ Container monitor_app Created
  ✓ Container prometheus_node_exporter Created
  ✓ Container app_two Created
  ✓ Container prometheus Created
  ✓ Container grafana Created
  ✓ Container postgres Created
  ✓ Container app_one Created
  ✓ Container postgres_exporter Created
```

The console logs indicating monitor\_app was deployed successfully.

|             |                         |                     |                                                           |                  |                  |          |            |            |               |  |  |
|-------------|-------------------------|---------------------|-----------------------------------------------------------|------------------|------------------|----------|------------|------------|---------------|--|--|
| monitor_app | 15:44:16                | - cmdstanpy         | - INFO                                                    | - Chain [1]      | start processing |          |            |            |               |  |  |
| monitor_app | 2024-09-25 15:44:16,050 | - INFO              | - Chain [1]                                               | start processing |                  |          |            |            |               |  |  |
| monitor_app | 15:44:16                | - cmdstanpy         | - INFO                                                    | - Chain [1]      | done processing  |          |            |            |               |  |  |
| monitor_app | 2024-09-25 15:44:16,148 | - INFO              | - Chain [1]                                               | done processing  |                  |          |            |            |               |  |  |
| monitor_app | 2024-09-25 15:44:16,213 | - INFO              | - Anomalies: 0, Cumulative: 26, MAE: 0.0913, MAPE: 5.8967 |                  |                  |          |            |            |               |  |  |
| monitor_app | 2024-09-25 15:44:16,216 | - INFO              | - All Results:                                            |                  |                  |          |            |            |               |  |  |
| monitor_app |                         | timestamp           |                                                           | ds               | y                | yhat     | yhat_lower | yhat_upper | anomaly_count |  |  |
| monitor_app | 2024-09-25 15:24:10     | 2024-09-25 15:23:10 | 0.686094                                                  | 0.251713         | 0.039757         | 0.466626 | 3          |            |               |  |  |
| monitor_app | 2024-09-25 15:24:10     | 2024-09-25 15:23:25 | 0.559208                                                  | 0.251713         | 0.029770         | 0.491097 | 3          |            |               |  |  |
| monitor_app | 2024-09-25 15:24:10     | 2024-09-25 15:23:40 | 0.058751                                                  | 0.251713         | 0.039555         | 0.465710 | 3          |            |               |  |  |
| monitor_app | 2024-09-25 15:24:10     | 2024-09-25 15:23:55 | 0.339113                                                  | 0.251713         | 0.025894         | 0.456737 | 3          |            |               |  |  |
| monitor_app | 2024-09-25 15:24:10     | 2024-09-25 15:24:10 | 0.536532                                                  | 0.251713         | 0.034437         | 0.462650 | 3          |            |               |  |  |
| monitor_app | 2024-09-25 15:26:10     | 2024-09-25 15:25:10 | 0.198536                                                  | 0.273264         | 0.109414         | 0.445998 | 5          |            |               |  |  |
| monitor_app | 2024-09-25 15:26:10     | 2024-09-25 15:25:25 | 0.010663                                                  | 0.273264         | 0.098315         | 0.438684 | 5          |            |               |  |  |
| monitor_app | 2024-09-25 15:26:10     | 2024-09-25 15:25:40 | 0.630423                                                  | 0.273264         | 0.095865         | 0.450981 | 5          |            |               |  |  |
| monitor_app | 2024-09-25 15:26:10     | 2024-09-25 15:25:55 | 0.198788                                                  | 0.273264         | 0.110128         | 0.454455 | 5          |            |               |  |  |
| monitor_app | 2024-09-25 15:26:10     | 2024-09-25 15:26:10 | 0.691354                                                  | 0.273264         | 0.090288         | 0.444832 | 5          |            |               |  |  |
| monitor_app | 2024-09-25 15:28:10     | 2024-09-25 15:27:10 | 0.814446                                                  | 0.276725         | 0.079553         | 0.480513 | 8          |            |               |  |  |
| monitor_app | 2024-09-25 15:28:10     | 2024-09-25 15:27:25 | 0.126532                                                  | 0.276725         | 0.086228         | 0.481626 | 8          |            |               |  |  |
| monitor_app | 2024-09-25 15:28:10     | 2024-09-25 15:27:40 | 0.304762                                                  | 0.276725         | 0.064232         | 0.478096 | 8          |            |               |  |  |
| monitor_app | 2024-09-25 15:28:10     | 2024-09-25 15:27:55 | 0.688664                                                  | 0.276725         | 0.072729         | 0.488325 | 8          |            |               |  |  |
| monitor_app | 2024-09-25 15:28:10     | 2024-09-25 15:28:10 | 0.772500                                                  | 0.276725         | 0.076455         | 0.484779 | 8          |            |               |  |  |
| monitor_app | 2024-09-25 15:30:13     | 2024-09-25 15:29:10 | 0.554092                                                  | 0.234596         | 0.008948         | 0.445189 | 11         |            |               |  |  |
| monitor_app | 2024-09-25 15:30:13     | 2024-09-25 15:29:25 | 0.567914                                                  | 0.234596         | 0.017829         | 0.441910 | 11         |            |               |  |  |

Continuous training of the model

|                       |             |                         |                     |                                                           |                  |                  |          |            |            |               |  |
|-----------------------|-------------|-------------------------|---------------------|-----------------------------------------------------------|------------------|------------------|----------|------------|------------|---------------|--|
| LAB2-Prometheus_Gr... | monitor_app | 13:49:43                | - cmdstanpy         | - INFO                                                    | - Chain [1]      | start processing |          |            |            |               |  |
| LAB3-Detecting...     | monitor_app | 2024-09-25 13:49:43,904 | - INFO              | - Chain [1]                                               | start processing |                  |          |            |            |               |  |
| > bin                 | monitor_app | 13:49:44                | - cmdstanpy         | - INFO                                                    | - Chain [1]      | done processing  |          |            |            |               |  |
| > containers          | monitor_app | 2024-09-25 13:49:44,011 | - INFO              | - Chain [1]                                               | done processing  |                  |          |            |            |               |  |
| > app_one             | monitor_app | 2024-09-25 13:49:44,037 | - INFO              | - Anomalies: 2, Cumulative: 28, MAE: 0.2537, MAPE: 0.6505 |                  |                  |          |            |            |               |  |
| > app_two             | monitor_app | 2024-09-25 13:49:44,041 | - INFO              | - All Results:                                            |                  |                  |          |            |            |               |  |
| > grafana             | monitor_app |                         | timestamp           |                                                           | ds               | y                | yhat     | yhat_lower | yhat_upper | anomaly_count |  |
| > monitor_app         | monitor_app | 2024-09-25 13:27:37     | 2024-09-25 13:26:36 | 0.606208                                                  | 0.370954         | 0.171582         | 0.573150 | 2          |            |               |  |
| > Dockerfile U        | monitor_app | 2024-09-25 13:27:37     | 2024-09-25 13:26:51 | 0.193013                                                  | 0.370954         | 0.176696         | 0.573289 | 2          |            |               |  |
| > requirement... U    | monitor_app | 2024-09-25 13:27:37     | 2024-09-25 13:27:06 | 0.336842                                                  | 0.370954         | 0.192552         | 0.565175 | 2          |            |               |  |
| ! config.yml U        | monitor_app | 2024-09-25 13:27:37     | 2024-09-25 13:27:21 | 0.680294                                                  | 0.370954         | 0.174531         | 0.553459 | 2          |            |               |  |
| > Dockerfile U        | monitor_app | 2024-09-25 13:27:37     | 2024-09-25 13:27:36 | 0.462552                                                  | 0.370954         | 0.167115         | 0.549195 | 2          |            |               |  |
| > docker-compo... U   | monitor_app | 2024-09-25 13:29:37     | 2024-09-25 13:28:37 | 0.517547                                                  | 0.306643         | 0.097535         | 0.523110 | 4          |            |               |  |
| READMEmd              | monitor_app | 2024-09-25 13:29:37     | 2024-09-25 13:28:52 | 0.304336                                                  | 0.306643         | 0.115358         | 0.506998 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:29:37     | 2024-09-25 13:29:07 | 0.245441                                                  | 0.306643         | 0.087802         | 0.522209 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:29:37     | 2024-09-25 13:29:22 | 0.669835                                                  | 0.306643         | 0.089031         | 0.501737 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:29:37     | 2024-09-25 13:29:37 | 0.791774                                                  | 0.306643         | 0.085753         | 0.520920 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:31:38     | 2024-09-25 13:30:37 | 0.155862                                                  | 0.310596         | 0.087264         | 0.519799 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:31:38     | 2024-09-25 13:30:52 | 0.390957                                                  | 0.310596         | 0.079180         | 0.537482 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:31:38     | 2024-09-25 13:31:07 | 0.504335                                                  | 0.310596         | 0.096436         | 0.539321 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:31:38     | 2024-09-25 13:31:22 | 0.361745                                                  | 0.310596         | 0.094767         | 0.519537 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:31:38     | 2024-09-25 13:31:37 | 0.395851                                                  | 0.310596         | 0.095674         | 0.538206 | 4          |            |               |  |
|                       | monitor_app | 2024-09-25 13:33:38     | 2024-09-25 13:32:38 | 0.300974                                                  | 0.320145         | 0.104777         | 0.543427 | 6          |            |               |  |
|                       | monitor_app | 2024-09-25 13:33:38     | 2024-09-25 13:32:53 | 0.008484                                                  | 0.320145         | 0.086561         | 0.544201 | 6          |            |               |  |
|                       | monitor_app | 2024-09-25 13:33:38     | 2024-09-25 13:33:08 | 0.655359                                                  | 0.320145         | 0.086626         | 0.544082 | 6          |            |               |  |
|                       | monitor_app | 2024-09-25 13:33:38     | 2024-09-25 13:33:23 | 0.968420                                                  | 0.320145         | 0.098417         | 0.537919 | 6          |            |               |  |
|                       | monitor_app | 2024-09-25 13:33:38     | 2024-09-25 13:33:38 | 0.186263                                                  | 0.320145         | 0.095174         | 0.537960 | 6          |            |               |  |
|                       | monitor_app | 2024-09-25 13:35:38     | 2024-09-25 13:34:38 | 0.567019                                                  | 0.346764         | 0.164499         | 0.533340 | 9          |            |               |  |
|                       | monitor_app | 2024-09-25 13:35:38     | 2024-09-25 13:34:53 | 0.605815                                                  | 0.346764         | 0.167773         | 0.546971 | 9          |            |               |  |
|                       | monitor_app | 2024-09-25 13:35:38     | 2024-09-25 13:35:08 | 0.311106                                                  | 0.346764         | 0.152954         | 0.533501 | 9          |            |               |  |
|                       | monitor_app | 2024-09-25 13:35:38     | 2024-09-25 13:35:23 | 0.226446                                                  | 0.346764         | 0.161971         | 0.538345 | 9          |            |               |  |
|                       | monitor_app | 2024-09-25 13:35:38     | 2024-09-25 13:35:38 | 0.894702                                                  | 0.346764         | 0.167434         | 0.542935 | 9          |            |               |  |
|                       | monitor_app | 2024-09-25 13:37:38     | 2024-09-25 13:36:38 | 0.613544                                                  | 0.370199         | 0.207153         | 0.554990 | 13         |            |               |  |
|                       | monitor_app | 2024-09-25 13:37:38     | 2024-09-25 13:36:53 | 0.351566                                                  | 0.370199         | 0.207084         | 0.556203 | 13         |            |               |  |
|                       | monitor_app | 2024-09-25 13:37:38     | 2024-09-25 13:37:08 | 0.972589                                                  | 0.370199         | 0.194858         | 0.556092 | 13         |            |               |  |
|                       | monitor_app | 2024-09-25 13:37:38     | 2024-09-25 13:37:23 | 0.747910                                                  | 0.370199         | 0.184297         | 0.569703 | 13         |            |               |  |
|                       | monitor_app | 2024-09-25 13:37:38     | 2024-09-25 13:37:38 | 0.965843                                                  | 0.370199         | 0.186101         | 0.546811 | 13         |            |               |  |

Indefinite training:

|                     |             |                     |                     |          |          |          |          |     |  |  |  |
|---------------------|-------------|---------------------|---------------------|----------|----------|----------|----------|-----|--|--|--|
| app.py U            | monitor_app | 2024-09-25 17:28:41 | 2024-09-25 17:28:10 | 0.035111 | 0.253486 | 0.039552 | 0.470582 | 154 |  |  |  |
| > Dockerfile U      | monitor_app | 2024-09-25 17:28:41 | 2024-09-25 17:28:25 | 0.533625 | 0.253486 | 0.063013 | 0.473371 | 154 |  |  |  |
| > app_two           | monitor_app | 2024-09-25 17:28:41 | 2024-09-25 17:28:40 | 0.698723 | 0.253486 | 0.047982 | 0.473581 | 154 |  |  |  |
| > grafana           | monitor_app | 2024-09-25 17:30:44 | 2024-09-25 17:29:41 | 0.769638 | 0.279789 | 0.044968 | 0.492573 | 157 |  |  |  |
| > monitor_app       | monitor_app | 2024-09-25 17:30:44 | 2024-09-25 17:29:56 | 0.969113 | 0.279789 | 0.036109 | 0.502486 | 157 |  |  |  |
| > prometheus        | monitor_app | 2024-09-25 17:30:44 | 2024-09-25 17:30:11 | 0.538842 | 0.279789 | 0.058448 | 0.503575 | 157 |  |  |  |
| > docker-compo... U | monitor_app | 2024-09-25 17:30:44 | 2024-09-25 17:30:26 | 0.359690 | 0.279789 | 0.046427 | 0.501893 | 157 |  |  |  |
| READMEmd            | monitor_app | 2024-09-25 17:30:44 | 2024-09-25 17:30:41 | 0.007215 | 0.279789 | 0.028485 | 0.514963 | 157 |  |  |  |
|                     | monitor_app | 2024-09-25 17:32:44 | 2024-09-25 17:31:44 | 0.695720 | 0.289730 | 0.042535 | 0.543698 | 159 |  |  |  |
|                     | monitor_app | 2024-09-25 17:32:44 | 2024-09-25 17:31:59 | 0.196397 | 0.289730 | 0.036201 | 0.534140 | 159 |  |  |  |
|                     | monitor_app | 2024-09-25 17:32:44 | 2024-09-25 17:32:14 | 0.219096 | 0.289730 | 0.040487 | 0.545100 | 159 |  |  |  |
|                     | monitor_app | 2024-09-25 17:32:44 | 2024-09-25 17:32:29 | 0.112152 | 0.289730 | 0.040126 | 0.539159 | 159 |  |  |  |
|                     | monitor_app | 2024-09-25 17:32:44 | 2024-09-25 17:32:44 | 0.531890 | 0.289730 | 0.061001 | 0.528286 | 159 |  |  |  |
|                     | monitor_app | 2024-09-26 07:42:32 | 2024-09-26 07:41:32 | 0.134253 | 0.226025 | 0.045946 | 0.419900 | 163 |  |  |  |
|                     | monitor_app | 2024-09-26 07:42:32 | 2024-09-26 07:41:47 | 0.461871 | 0.226025 | 0.042279 | 0.410457 | 163 |  |  |  |
|                     | monitor_app | 2024-09-26 07:42:32 | 2024-09-26 07:42:02 | 0.686743 | 0.226025 | 0.043477 | 0.396607 | 163 |  |  |  |
|                     | monitor_app | 2024-09-26 07:42:32 | 2024-09-26 07:42:17 | 0.712090 | 0.226025 | 0.038931 | 0.405897 | 163 |  |  |  |
|                     | monitor_app | 2024-09-26 07:42:32 | 2024-09-26 07:42:32 | 0.762867 | 0.226025 | 0.049925 | 0.415769 | 163 |  |  |  |
|                     | monitor_app | 2024-09-26 07:44:33 | 2024-09-26 07:43:32 | 0.860747 | 0.226473 | 0.039734 | 0.414279 | 167 |  |  |  |
|                     | monitor_app | 2024-09-26 07:44:33 | 2024-09-26 07:43:47 | 0.764978 | 0.226473 | 0.011280 | 0.418144 | 167 |  |  |  |
|                     | monitor_app | 2024-09-26 07:44:33 | 2024-09-26 07:44:02 | 0.065790 | 0.226473 | 0.032965 | 0.429318 | 167 |  |  |  |
|                     | monitor_app | 2024-09-26 07:44:33 | 2024-09-26 07:44:17 | 0.958722 | 0.226473 | 0.022376 | 0.417665 | 167 |  |  |  |
|                     | monitor_app | 2024-09-26 07:44:33 | 2024-09-26 07:44:32 | 0.836130 | 0.226473 | 0.038204 | 0.420440 | 167 |  |  |  |

## Legend:

**timestamp:** This column represents the time at which the result was generated or recorded.

**ds:** This column stands for "datestamp" and represents the time of the observed data points in the test dataset.

**y:** This column contains the actual observed values of the metric from the test data.

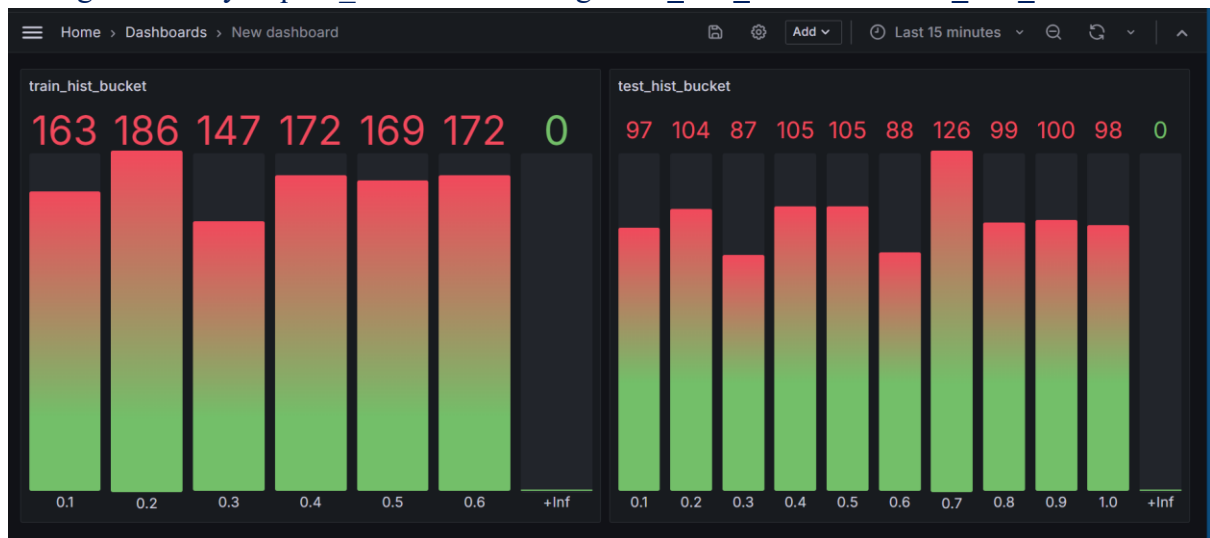
**yhat:** This is the forecasted value for the metric generated by the Prophet model. It's the predicted value based on the trained model, for the given ds time point.

**yhat\_lower:** This represents the lower bound of the predicted confidence interval.

**yhat\_upper:** This represents the upper bound of the predicted confidence interval.

**anomaly\_count:** This column keeps track of the cumulative number of anomalies detected up to that point in time.

## Histograms of my request\_time metrics using **train\_hist\_bucket** and **test\_hist\_bucket**



## Grafana dashboard of the **train\_gauge**, **test\_gauge**, **mape\_metric**, **anomaly\_count** and **mae\_metric** in 5 minutes timeframe



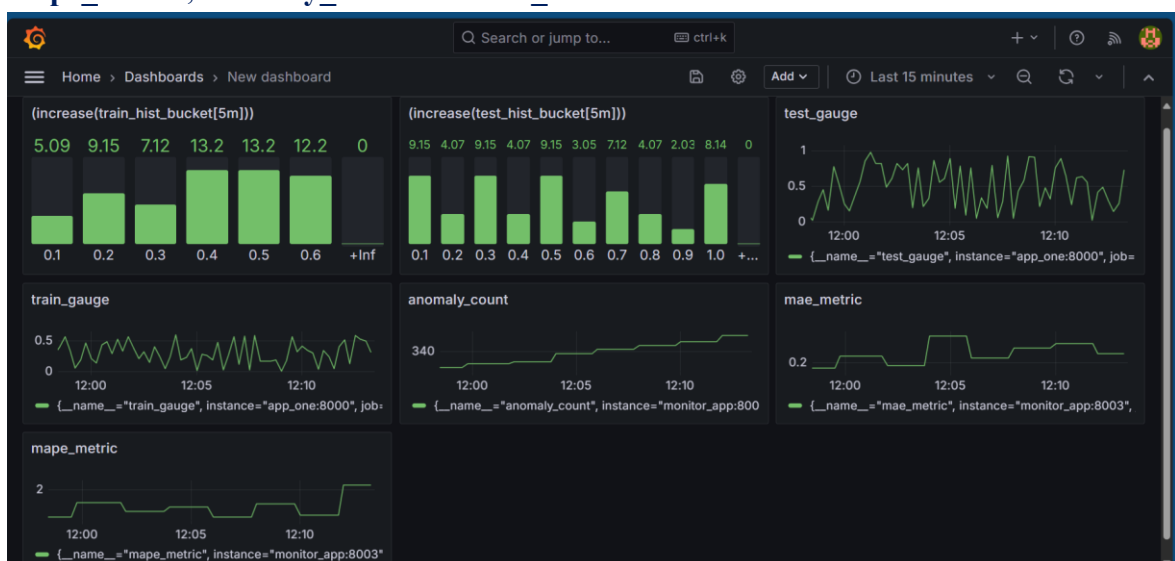
Grafana dashboard of the **train\_guage**, **test\_guage** , **mape\_metric**, **anomaly\_count** and **mae\_metric** in **10 minutes** timeframe



Grafana dashboard of the **train\_guage**, **test\_guage** , **mape\_metric**, **anomaly\_count** and **mae\_metric** in **15 minutes** timeframe



Grafana dashboard of the **train\_hist\_bucket**, **test\_hist\_bucket** train\_guage, test\_guage , **mape\_metric**, **anomaly\_count** and **mae\_metric** in **15 minutes** timeframe



Lab Task – Answer the following questions.

### **1. Reasonable Baseline of Data for Prophet Model**

A reasonable baseline of data for using a Prophet model in a production system depends on the frequency and nature of the data. For time series forecasting, it's recommended to have at least one year of historical data to capture seasonal patterns and trends effectively. However, if the data is more granular (e.g., minute-level data), a few months of data might suffice.

#### **(a) Considerations:**

- Prophet is designed to handle seasonality and trends. Having at least one full cycle of seasonal data (e.g., one year for yearly seasonality) helps the model learn these patterns.
- For high-frequency data (e.g., minute-level), a few months might be enough, but for daily or weekly data, a longer history is beneficial.
- Ensuring the data is clean and consistent. Missing values and outliers should be handled appropriately.

#### **(b) Operational Challenges:**

- Large datasets can lead to increased computational time and resource usage. This might require more powerful hardware or cloud resources.
- Depending on the data size and frequency, training the model can take significant time. This might delay the deployment of the model or updates.
- Storing large volumes of historical data can be challenging and might require efficient data storage solutions.

### **2. Continuous Retraining vs. Manual Review/Approval**

#### **Continuous Retraining:**

##### **Advantage:**

- The model can quickly adapt to new patterns and changes in the data, improving its accuracy over time.
- Reduces the need for manual intervention, saving time and resources.

##### **Disadvantage:**

- Continuous retraining without proper validation can lead to overfitting, where the model performs well on training data but poorly on unseen data.
- Frequent retraining can be computationally expensive and resource intensive.
- The model might learn from anomalies or noise in the data, leading to inaccurate predictions.

#### **Manual Review/Approval:**

##### **Advantage:**

Quality Control: Allows for human oversight to ensure the model is performing as expected and not learning from anomalies or noise.

Stability: Reduces the risk of overfitting and ensures the model is only updated when necessary.

##### **Disadvantage:**

- Manual review can introduce delays in updating the model, potentially reducing its adaptability to new patterns.
- Requires human resources for review and approval, which can be time-consuming and costly.

### **Disadvantage of Fully Automatic Operation:**

- Without manual review, there is a risk of the model learning from incorrect or anomalous data, leading to poor performance.
- Continuous retraining without proper validation can cause the model to overfit to recent data, reducing its generalizability.
- Frequent retraining can be resource-intensive, requiring significant computational power and storage.

### **Recommendations**

- Implementing a hybrid approach where the model retrains periodically (e.g., weekly, or monthly) with manual review and approval for significant updates.
- Continuously monitoring the model's performance using metrics like MAE and MAPE. Setting up alerts for significant deviations.
- Using a robust validation strategy to ensure the model is not overfitting and performs well on unseen data.
- Ensuring efficient data storage and management practices to handle large volumes of historical data.